

**WINE CONSUMPTION AND ITS CORRELATION WITH
CONSUMERS' INCOME**

Nguyen Dinh An Hanh

Fulbright University Vietnam

MATH105 - Introduction to Data Analysis

Instructor: Nguyen Minh Quan

April 26, 2022

I. ABSTRACT

Previous research investigated the studies of the correlation between wine consumption and income methodologically. Some researchers have discovered that alcohol use decreases earnings, while others have found it has no effect or raises incomes. Thus, this study aims to examine the relationship between alcohol intake and other demographic characteristics, especially earnings in order to resolve contradictions and establish their relationship. The statistical method used in the paper: are heat map, hypothesis testing, and linear regression. Results: Among four variables (education, age, marital status), income is proven to be the most affected variable on wine consumption, which positively correlates together. Another finding is that toxic drinkers earn less than moderate drinkers, which contradicts our previous finding.

II. INTRODUCTION

The correlation between wine consumption and earnings has been a long-running topic that has resulted in a host of concerns. It is quantitatively and qualitatively evident that the over-consumption of wine could lead to elements of risk, including income change or lifespan decline. While academia is paying more attention to the correlation between alcohol or other substance addiction and wages, multiple pieces of research show differences in their results and analysis. Mullahy and Sindelar (1991) demonstrated that wine consumption harms income. Benham's findings in 1982 also share the same results. Contrastingly, in 1998, Berger and Leigh's proved that individuals who drink earn significantly higher than those who do not.

These analyses have achieved partially contradicting conclusions. Therefore, this report will examine the relationship between wine consumption and income. Similar to prior literature, drinkers are categorized into two types, moderate and heavy ones. However, we believe other confounding variables generate false or misleading data between the two main

variables. Therefore, age, marital status, and education impact would be assessed with a heatmap.

III. METHODOLOGY

Dr. Omar Romero-Hernandez (2021) provides the dataset for this project via a Kaggle page. It is licensed as CC0: Public Domain, which states, “You can copy, modify, distribute and perform the work, even for commercial purposes, all without asking permission.” The sample size is 2240.

The analysis process is conducted via Google Colab, by programming language Python.

1. Data Processing

To analyze the correlation between the wine consumption of a person and their income and whether it affects their amount spent on wine, we extract five columns of the dataset: Year_Birth (age), Education (education qualification), Marital_Status (marital status), Income (yearly household income) and MtnWines (amount spent on wine).

Other unnecessary data, including either row of null values or outliers, are also removed from the data frame for accuracy. The column on marital status contains two jamming data named “Absurd” and “YOLO,” which clearly do not describe a person’s status. Those data will be filtered from the dataset. For the outliers clearing process, data within the lower and upper whisker are selected.

The categorical data namely marital status and educational qualification are converted to discrete variables. Accordingly, the lowest education qualification is listed as “1” and increases as the degree level to the maximum of “5”. Similarly, marital status is numbered from the first to the last stage of each relationship situation.

Basic	Graduation	2n Cycle	Master	PhD
1	2	3	4	5

Single	Alone	Together	Married	Divorced	Widow
6	7	8	9	10	11

Table 1. Categorical data converted to discrete data.

2. Methods

2.1 Heatmap

In this report, variables that significantly affect wine consumption are expected to be detected. Using a heat map, the corresponding coefficient among all variables is summarized in one table and represents their significance by the intensity level of color. By doing so, the most and least affected variables can be identified quickly.

2.2 Hypothesis testing

Consequently, a hypothesis testing will be constructed to evaluate whether there is convincing evidence to statistically differentiate between the income earned by moderate and toxic drinkers. Regarding differences in their mean incomes, a one-tailed hypothesis testing is applied.

$$\text{Hypothesis: } H_0: \mu_{toxic} \geq \mu_{moderate} \text{ vs. } H_A: \mu_{toxic} < \mu_{moderate}$$

The P-value is computed according to the significance level of 5% in order to accept or reject the null hypothesis. If the P-value exceeds the significance level, the null-hypothesis is failed to reject.

2.3 Regression Analysis

A linear regression is constructed to put the variables in a broad context. Wine consumption of each individual is hypothesized to depend upon a vector X_i of annual income, age, education qualification, and marital status.

$$W = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

The ordinal Least Squared Method (OLS) was also used to test the hypothesized relationship. Firstly, we set up the hypothesis. Then we will calculate the p-value by computing program to conclude our result.

- **H₀:** $\beta_i = 0$ (variable X has no influence on X)
- **H_A:** $\beta_i \neq 0$ (variable X has a significant influence on Y)

IV. RESULTS

1. Descriptive statistics (10pts):

This study, directing at testing the hypothesis, starts with an overview of data. The means, media and standard deviation of numerical variables is showed in table 2.

Table 2. Variables and descriptive statistics

	Variables	Mean	S.D	Median	Min	Max
1	WINE CONSUMPTION (\$)	306.11	337.78	176	0	1224
2	INCOME (\$)	51606.14	20721.18	51287	1730	113734

3	EDUCATION	***	***	***	***	***
4	AGE	53.1	11.7	52	26	82
5	MARITAL STATUS	***	***	***	***	***

To examine the spread and the normality of investigated numerical data, we calculate the skewness (S_{KP}), then utilize computer programming to visualize how variables might vary. The probability density curves of wine consumption and income are depicted in Figures 1 and 2, respectively. Although we have removed the significant outliers, the amount of spending on wine still experienced a sharp left-skewed in variables. ($S_{KP} = 1.048 > 1$). Income distribution, on the other hand, seems to have a more normal and symmetrical shape ($-0.5 < S_{KP} = 0.0269 < 0.5$).

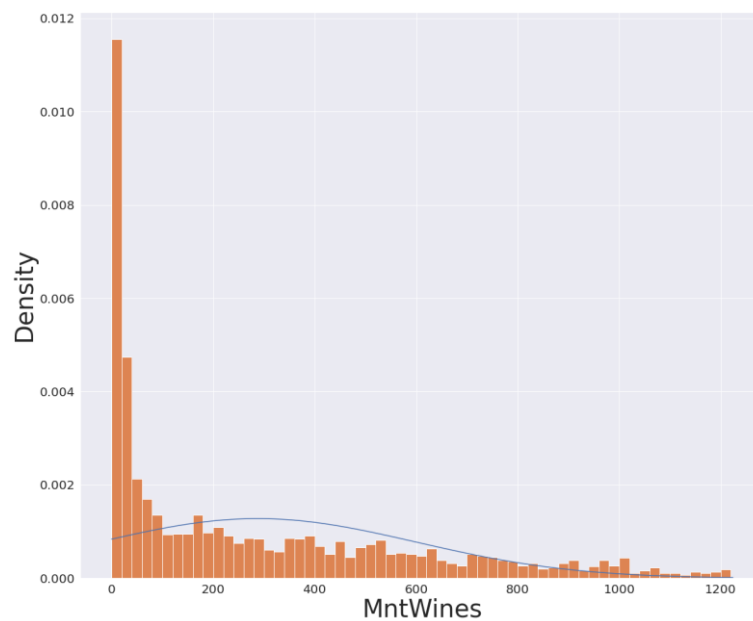


Figure 1. Probability density curve of Wine Consumption

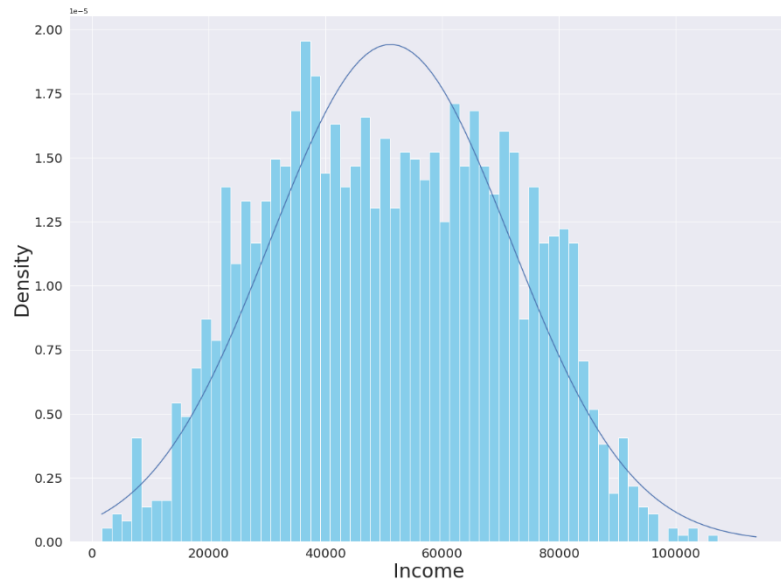


Figure 2. Probability density curve of Income

To turn to categorical data, a histogram is used to investigate the spread of the data set. (Figure 3 and 4 below).

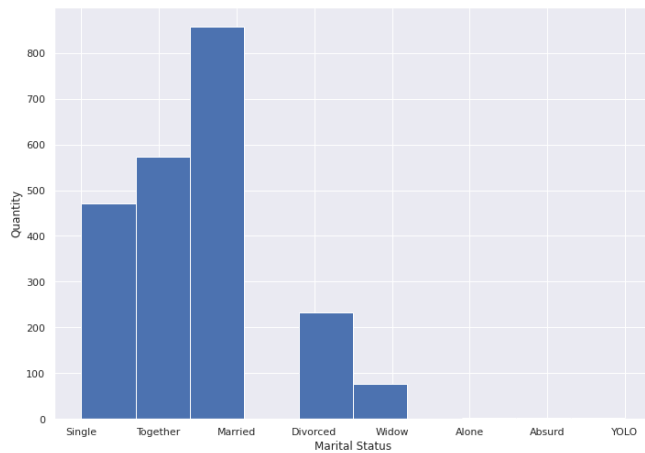


Figure 3. Histogram of Marital Status

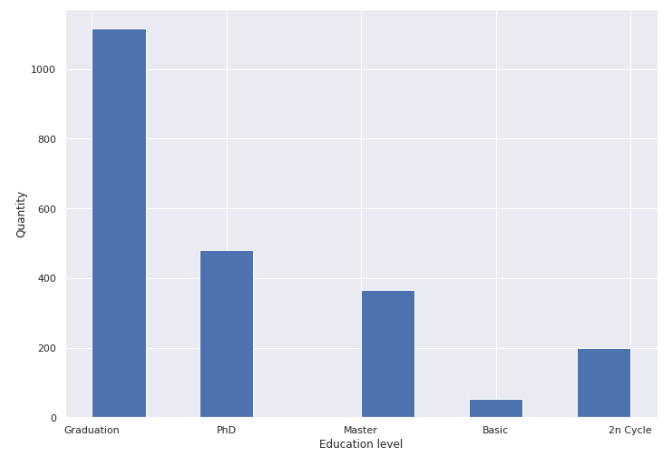


Figure 4. Histogram of Educational level

2. Inferential Statistics:

2.1 RESULT 1: There is an association between wine consumption and other demographics.

The ordinal least squared was used to perform the hypothesized testing since it is the most common and simple statistical approach when conducting correlation testing. This Regression Model allows researchers to examine the specific effect of each independent variable on the dependent variable. Table 2 presents the OLS result. However, we should focus on the coefficient and p-value.

Table 3. OLS Result

<i>Characteristics</i>	<i>Coefficient</i>	<i>Standard error</i>	<i>t</i>	<i>p-value</i>
WINE CONSUMPTION				
Income	432.4373	11.295	38.284	0.000
Edu	21.3302	4.343	4.912	0.000
Age	0.0147	0.49	0.03	0.976
Marital Status	-0.1705	2.23	-0.076	0.939
CONSTANT	- 4406.81	117.374		
NO. OBSERVATIONS	2166			
R^2	0.430			
F-STATISTIC	414.1			

**The dummy variables are also included in the test since it have been transformed it into numerical data.*

The table indicates a positive correlation between wine consumption and income ($\beta_1 = 432.4373$), wine consumption and education ($\beta_1 = 21.3302$). The p-value for both is $0 < 0.05$, which helps us reject the null hypothesis (X have no influence

on Y) and re-confirm their correlation. While for age and marital status, the p-value is bigger than 0.05: 0.976 and 0.939 respectively, thus, we accept the null hypothesis.

2.2 RESULT 2: Wine consumption and income have the strongest correlation compared to other demographics.

After confirming the relationship between wine consumption and other demographics, we performed a heatmap to visualize the correlations of all variables and consider which variable has the strongest correlation with *the consumption of wines*. Table 2 demonstrated the correlation coefficient r , a statistics variable, indicates the strength and direction of a linear relationship.

Table 2. The correlation coefficient r between wines consumption and different variables

Indicator		1	2	3	4	5
1	WINE CONSUMPTION	1				
2	INCOME (\$)	0.734	1			
3	EDUCATION	0.153	0.108	1		
4	AGE	0.162	0.212	0.155	1	
5	MARITAL STATUS	0.033	0.338	0.04	0.155	1

As illustrated by the table, wine consumption and income have the strongest relationship ($r = 0.73$). While there appears to be no association between the three variables (education, age, marital status) and wine consumption since the Pearson r is too insignificant (0.153, 0.162, 0.033 respectively).

2.3 RESULT 3: Toxic drinkers earn more than moderate drinkers.

To investigate more profoundly, we utilize left-tailed hypothesis testing to reassess the characteristic of the correlation to see whether it is positive or negative.

Firstly, we separate the wine consumers into two groups by comparing them with the median: toxic drinkers (amount of money spend on wine higher than 176\$), and moderate drinkers (amount of money spend on wine lower than 176\$). In our comparison of mean wine consumption between toxic and moderate drinkers, we found an average difference of 587\$ and a p-value of 0.049; thus, we reject the null hypothesis that toxic drinkers earn more than moderate drinkers and conclude that there is likely a difference in earnings between them.

V. DISCUSSIONS

Strength: The method that this research use is hypothesis testing which provides a reliable framework for any data decisions. It proves whether the data is statistically significant and it is a valid method. Also, the heatmap shows a clearer correlation between variables that contribute to the hypothesis.

Limitations: The limitation of this research is the hypothesis testing has a result that counter the other two tests. Also, there is no distinction between the two type of drinkers when using the heatmap which does not create the distinction between the correlation of variables. Because the data set was taken from a company over its customer consumption, there are limitations to the data as it is not as vast as it could be. Thus giving the unexpected result against the trend. Moreover, the data set of wine consumption is not normal, thus it might effect the research's result.

VI. TEAMMATE EVALUATION

Link:

https://docs.google.com/document/d/1zHRmAmYhdupWHky0bmcgy_oR2DCGGzI1ZF80kkAVets/edit?usp=sharing

REFERENCES

- Berger, M. C., & Leigh, J. P. (1988). The effect of alcohol use on wages. *Applied Economics*, 20(10), 1343–1351. <https://doi.org/10.1080/00036848800000105>
- Mullahy, J., & Sindelar, J. L. (1991). Gender Differences in Labor Market Effects of Alcoholism. *American Economic Review*, 81(2), 161–165.
https://econpapers.repec.org/article/aeaaecrev/v_3a81_3ay_3a1991_3ai_3a2_3ap_3a161-65.htm
- Romero-Hernandez, O. (2021). *Customer Personality Analysis*. marketing_campaign.csv.
https://raw.githubusercontent.com/amankharwal/Website-data/master/marketing_campaign.csv